

Detecting AI-Generated Text

SEA 820 Natural Language Processing Final Project

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GitHub: <https://github.com/Atem2222/SEA-820-NLP-Final-Project-Detecting-AI-Generated-Text>

Abstract

This project evaluated two approaches for detecting AI-generated text: TF-IDF with Logistic Regression and a fine-tuned DistilBERT classifier. A balanced sample of 40,000 texts was divided into training, validation, and test sets using a stratified 80/10/10 split. The classic baseline achieved 98.67% accuracy and 98.67% F1-score. DistilBERT achieved 99.30% accuracy and 99.30% F1-score. The Transformer reduced false negatives from 40 to 12, although false positives increased from 13 to 16. Detailed error analysis showed that formulaic human essays could be flagged as AI, while generated texts containing deliberate spelling and grammar errors could evade detection. The results demonstrate strong in-domain performance but also show that AI-text detectors should be used only as screening tools with human review.

1. Introduction

Large language models can generate fluent essays, articles, and responses at low cost. This creates challenges for academic integrity, misinformation detection, and online moderation. AI-generated text detection is therefore an important classification problem, but incorrect predictions can have serious consequences. A false positive may wrongly accuse a human author, while a false negative may allow generated content to pass undetected.

The project objective was to compare a transparent classic machine-learning baseline against a contextual Transformer model. The study focused not only on overall accuracy, but also on precision, recall, F1-score, confusion matrices, model-specific mistakes, text-length effects, and ethical implications.

2. Dataset and Preparation

The project used an AI vs. Human Text dataset from Kaggle. Because the original dataset contained more than 360,000 records, a balanced sample of 40,000 texts was created: 20,000 human-written and 20,000 AI-generated. The final sample contained no missing values and no duplicate texts.

Class	Label	Count
Human-written	0	20,000
AI-generated	1	20,000

Human-written texts averaged 422.1 words, while AI-generated texts averaged 344.1 words. This 78word difference was treated as a possible dataset shortcut. The data was split into 32,000 training texts, 4,000 validation texts, and 4,000 test texts using stratification and random seed 42.

3. Methods

3.1 TF-IDF and Logistic Regression

The baseline transformed documents into TF-IDF vectors using unigrams and bigrams, English stopword removal, sublinear term frequency, minimum document frequency of three, and a maximum vocabulary of 50,000 features. Logistic Regression was then trained on the sparse feature matrix.

3.2 DistilBERT Fine-Tuning

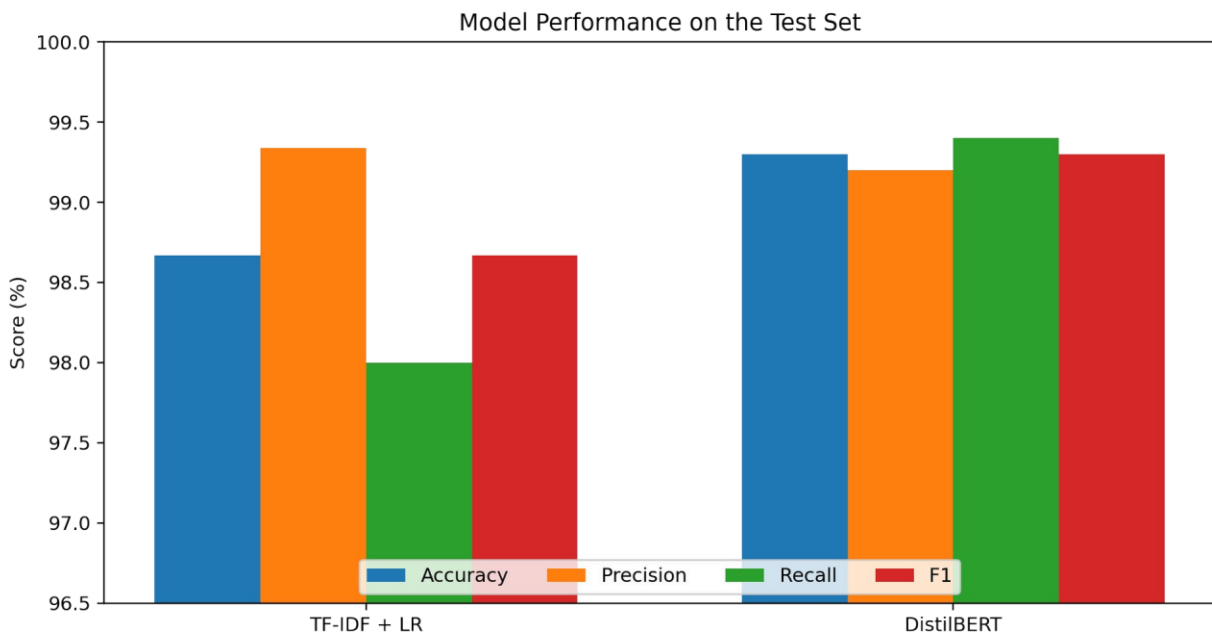
The Transformer model used distilbert-base-uncased with dynamic padding and a maximum sequence length of 256 tokens. Two controlled experiments compared learning rates of $2e-5$ and $3e-5$ and training batch sizes of 8 and 16. Each configuration trained for up to two epochs. Validation F1-score selected the best checkpoint, and the untouched test set was used only for final evaluation.

3.3 Evaluation

Both models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Precision and recall were calculated for the AI-generated class. The analysis also inspected highly confident false positives and false negatives, shared errors, model-specific errors, influential TF-IDF features, and error rates by text-length quartile.

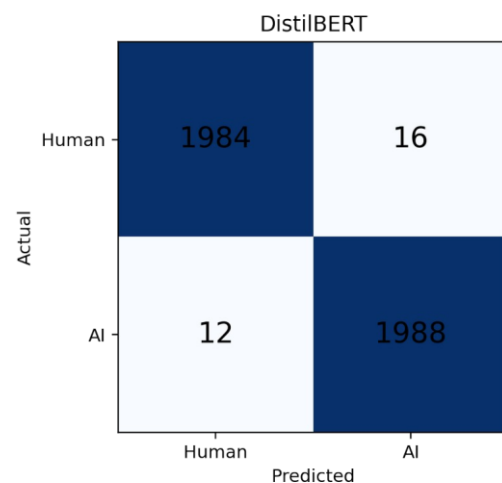
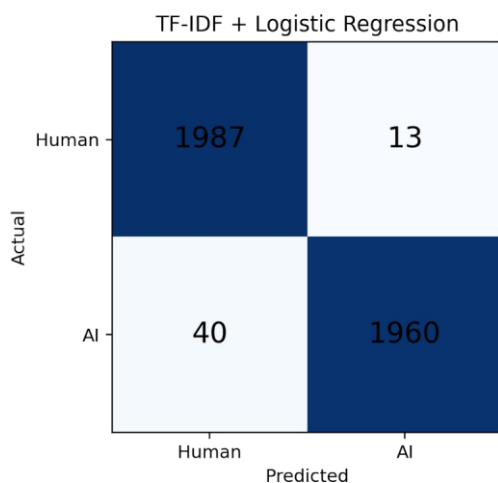
4. Results

Model	Accuracy	Precision	Recall	F1-score
TF-IDF + Logistic Regression	98.67%	99.34%	98.00%	98.67%
DistilBERT	99.30%	99.20%	99.40%	99.30%



DistilBERT improved accuracy by 0.62 percentage points and F1-score by 0.63 percentage points. The largest improvement was recall, which increased by 1.40 percentage points. Precision decreased slightly by 0.14 percentage points.

4.1 Confusion Matrices



The baseline correctly classified 1,987 human texts and 1,960 AI texts, with 13 false positives and 40 false negatives. DistilBERT correctly classified 1,984 human texts and 1,988 AI texts, with 16 false positives and 12 false negatives. The Transformer detected 28 additional AI texts, but incorrectly flagged three additional human texts.

5. Error Analysis

The two models shared 12 errors. DistilBERT corrected 41 baseline mistakes but introduced 16 new errors on texts that Logistic Regression classified correctly. The baseline performed worst on the shortest text quartile, while DistilBERT performed worst on the longest quartile, which may partly reflect the 256token truncation limit.

5.1 False Positives

Several human-written essays were flagged because they used clear argumentative structures, repeated transitions, and formulaic conclusions. One 537-word human essay received an AI probability of 0.9997. Its organized structure and repeated phrases such as "One of the main reasons" and "Another reason" outweighed visible grammar errors. This example demonstrates that high confidence is not proof of AI authorship.

5.2 False Negatives

Generated texts were sometimes classified as human when they contained deliberate spelling errors, awkward grammar, placeholder names, or corrupted words. Examples included phrases such as "I thing," "Lometimes," and repeated character substitutions. These outputs resembled noisy student writing and were difficult for both lexical and contextual models.

5.3 Feature Analysis

The baseline associated words such as "important," "additionally," "potential," "essay," "conclusion," "firstly," and "essential" with AI-generated text. Human-associated terms included "people," "going," "car," "students," "percent," "paragraph," and "school." These correlations may reflect topics or dataset sources rather than universal differences between human and AI writing.

6. Ethical Considerations

False positives are the most serious deployment risk because a human author could be wrongly accused. The baseline produced 13 false positives and DistilBERT produced 16. Formal organization and learned academic templates may particularly affect non-native English writers, who may use repeated transitions, translation tools, or grammar support.

False negatives also show that generated text can evade detection through editing, paraphrasing, and deliberately introduced errors. The detector should therefore be used only as a screening tool. Any flagged work should be reviewed using drafts, writing history, citations, assignment context, and an opportunity for the author to explain the writing process.

7. Limitations

- The dataset was balanced, while real-world class proportions may differ.
- The models may learn topic, source, or formatting artifacts.
- DistilBERT truncated texts beyond 256 tokens.
- Only one Transformer architecture was evaluated.
- Performance on newer language models and edited AI text is unknown.
- High confidence did not guarantee correctness.

8. Conclusion

DistilBERT was the stronger overall model, achieving 99.30% F1-score compared with 98.67% for TFIDF + Logistic Regression. It substantially reduced false negatives, but slightly increased false positives. The detailed errors showed that formal human writing may be mistaken for AI and that generated text with intentional errors may appear human. The project therefore supports cautious, human-in-the-loop use rather than automatic decisions.

References

1. Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding.
2. Sanh, V., Debut, L., Chaumond, J., and Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter.
3. Wolf, T. et al. (2020). Transformers: State-of-the-Art Natural Language Processing.
4. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python.
5. Kaggle. AI vs. Human Text Balanced Dataset. <https://www.kaggle.com/datasets/arjunverma2004/aivs-human-text-balanced-180k-records?resource=download>

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